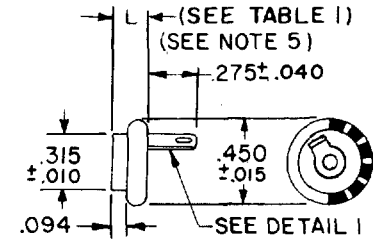
MIL-C-10950/3E
8 July 1968
SUPERSEDING
MIL-C-10950/3D
7 October 1963

MILITARY SPECIFICATION SHEET

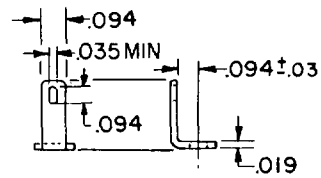
# CAPACITORS, FIXED, MICA DIELECTRIC, BUTTON STYLE (STANDOFF) STYLE CB31

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-10950.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

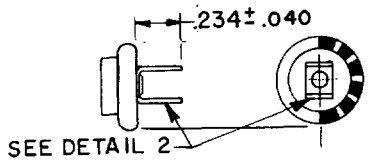



WITH TERMINAL ASSEMBLY P AND SHELL DIMENSIONS

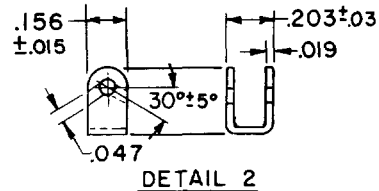



DETAIL I

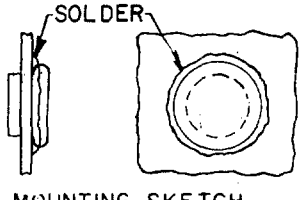
| INCHES | MM |
|---|---|
| .005 | .13 |
| .010 | .25 |
| .015 | .38 |
| .019 | .48 |
| .031 | .79 |
| .035 | .89 |
| .040 | 1.02 |
| .047 | 1.19 |
| .094 | 2.39 |
| .156 | 3.96 |
| .179 | 4.55 |
| .203 | 5.16 |
| .234 | 5.94 |
| .275 | 6.99 |
| .315 | 8.00 |
| .450 | 11.43 |



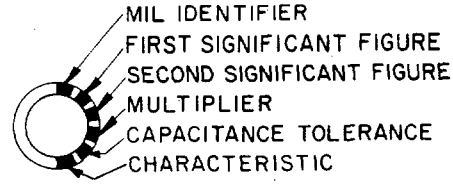

WITH TERMINAL ASSEMBLY Q




DETAIL 2




MOUNTING SKETCH




COLOR-CODE MARKING

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is .xxx±.005 (.13 mm).
4. Capacitors shall be soldered securely to mounting plate at as many points as necessary.
5. All dimensions are exclusive of sealing compound and color code.
6. Color-coded section shall not exceed 240°.



FSC 5910

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TABLE I. Style CB31.

| Part number | Capacitance | DC rated voltage | L 2/ Dimension | Characteristics available at capacitance tolerance | | | | Terminal assemblies |
|---|---|---|---|---|---|---|---|---|
| | | | | ±1% (F) | ±2% (G) | ±5% (J) | ±10% (K) | |
| | pf | Volts | | | | | | |
| CB31-B050K | 5 | 500 | 0.179 | | | | B | P, Q |
| CB31-D100K | 10 | 500 | 0.179 | | | | D | P, Q |
| CB31-D150K | 15 | 500 | 0.179 | | | | D | P, Q |
| CB31-D180K | 18 | 500 | 0.179 | | | | D | P, Q |
| CB31-D200J | 20 | 500 | 0.179 | | | D | - | P, Q |
| CB31-D220- | 22 | 500 | 0.179 | | | D | D | P, Q |
| CB31-D240J | 24 | 500 | 0.179 | | | D | - | P, Q |
| CB31-D270- | 27 | 500 | 0.179 | | | D | D | P, Q |
| CB31-D300J | 30 | 500 | 0.179 | | | D | - | P, Q |
| CB31-D330- | 33 | 500 | 0.179 | | | D | D | P, Q |
| CB31-D390- | 39 | 500 | 0.179 | | | D | D | P, Q |
| CB31-D470- | 47 | 500 | 0.179 | | | D | D | P, Q |
| CB31--510- | 51 | 500 | 0.179 | D, E | D, E | D, E | - | P, Q |
| CB31--560- | 56 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--620- | 62 | 500 | 0.179 | D, E | D, E | D, E | - | P, Q |
| CB31--680- | 68 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--750- | 75 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--101- | 100 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--111- | 110 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--151- | 150 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--161- | 160 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--201- | 200 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--221- | 220 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--241- | 240 | 500 | 0.179 | D, E | D, E | D, E | - | P, Q |
| CB31--271- | 270 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--301- | 300 | 500 | 0.179 | D, E | D, E | D, E | - | P, Q |
| CB31--331- | 330 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--361- | 360 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--431- | 430 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--471- | 470 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--511- | 510 | 500 | 0.179 | D, E | D, E | D, E | - | P, Q |
| CB31--561- | 560 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--621- | 620 | 500 | 0.179 | D, E | D, E | D, E | - | P, Q |
| CB31--681- | 680 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--751- | 750 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--102- | 1,000 | 500 | 0.179 | D, E | D, E | D, E | D | P, Q |
| CB31--152- | 1,500 | 500 | 0.234 | D, E | D, E | D, E | D | P, Q |
| CB31--182- | 1,800 | 500 | 0.234 | D, E | D, E | D, E | D | P, Q |
| CB31--222- | 2,200 | 500 | 0.234 | D, E | D, E | D, E | D | P, Q |
| CB31--242- | 2,400 | 500 | 0.234 | D, E | D, E | D, E | D | P, Q |

1/ Complete part number will include additional symbols to indicate terminal assembly, characteristic, and capacitance tolerance.

2/ Tolerances for L dimension shall be as follows:

0.179 ± 0.031
0.234 ± 0.040



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REQUIREMENTS:

Dimensions and configuration:

Case: See figure and table I.

Terminals: See figure.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: 500 volts.

Dissipation factor: See figure 1 of MIL-C-10950.

Characteristics: See table I.

| Symbol | Temperature coefficient | Capacitance drift |
|---|---|---|
| B | Not specified | Not specified |
| D | -100 to +100 | 0.3 percent or 0.3 pf whichever is greater. |
| E | -20 to +100 | ± (0.1 percent + 0.1 pf). |

Operating temperature range:
-55° C to 85° C.

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
200 percent of rated dc voltage for not less than 1 second, nor more than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202, condition C (70,000 ft.)
Test potential - 125 percent of rated dc voltage for not less than 1 second nor more than 5 seconds. Surge current shall not exceed 5 ma.

Insulation resistance:
Method 302 of MIL-STD-202, condition A or B.
At 25° C - 50,000 megohms minimum.
At 85° C - 15,000 megohms minimum.

Capacitance:
Method 305 of MIL-STD-202.
Measurement frequency
Capacitance values of 1000 pf or less - 1 MHz ±100 kHz.
Capacitance values of greater than 1000 pf - 1 kHz ±100 Hz.

Shock (medium impact):
Method 205 of MIL-STD-202, condition C (50G).

Vibration:
Method 204 of MIL-STD-202, condition B.

Moisture resistance:
Test method 106 of MIL-STD-202.
Insulation resistance:
Not less than 500 megohms.
Capacitance change:
Within 1 percent of the initial value or 0.5 pf, whichever is greater.
DF - 200 percent of initial requirement.

Temperature cycling and immersion:
Temperature cycling:
Test method 102 of MIL-STD-202, condition D.
Immersion:
Test method 104 of MIL-STD-202, condition A.
Insulation resistance:
Not less than 5000 megohms.
Capacitance change:
Within 1 percent of the initial value or 0.5 pf, whichever is greater.

Life:

2000 ±12 hours at $85^{\circ}\ ^{+3^{\circ}}_{-0^{\circ}}$ C.

Insulation resistance:
Greater than 50,000 megohms.
Capacitance change:
Within 3 percent of the initial value or 0.5 pf, whichever is greater.

Temperature coefficient and capacitance drift:
Within the limits specified for the applicable characteristic.

Torque:
In accordance with MIL-C-10950.
Mounting - normal mounting means.
Load - 4 ounce-inches.

NOTES:
1. Conversion of dash numbers to part numbers is as shown in table II.
2. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.



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TABLE II. Dash number to part number conversion.

| Dash number in MIL-C-10950/3D | | | | Converted to part number 1/ in MIL-C-10950/3E |
|---|---|---|---|---|
| Characteristic D | | Characteristic E | | |
| Terminal assembly | | | | |
| P | Q | P | Q | |
| 0001 | 0002 | 0003 | 0004 | CB31--510F |
| 0005 | 0006 | 0007 | 0008 | CB31--560F |
| 0009 | 0010 | 0011 | 0012 | CB31--620F |
| 0013 | 0014 | 0015 | 0016 | CB31--680F |
| 0017 | 0018 | 0019 | 0020 | CB31--750F |
| 0021 | 0022 | 0023 | 0024 | CB31--101F |
| 0025 | 0026 | 0027 | 0028 | CB31--111F |
| 0029 | 0030 | 0031 | 0032 | CB31--151F |
| 0033 | 0034 | 0035 | 0036 | CB31--161F |
| 0037 | 0038 | 0039 | 0040 | CB31--201F |
| 0041 | 0042 | 0043 | 0044 | CB31--221F |
| 0045 | 0046 | 0047 | 0048 | CB31--241F |
| 0049 | 0050 | 0051 | 0052 | CB31--271F |
| 0053 | 0054 | 0055 | 0056 | CB31--301F |
| 0057 | 0058 | 0059 | 0060 | CB31--331F |
| 0061 | 0062 | 0063 | 0064 | CB31--361F |
| 0065 | 0066 | 0067 | 0068 | CB31--431F |
| 0069 | 0070 | 0071 | 0072 | CB31--471F |
| 0073 | 0074 | 0075 | 0076 | CB31--511F |
| 0077 | 0078 | 0079 | 0080 | CB31--561F |
| 0081 | 0082 | 0083 | 0084 | CB31--621F |
| 0085 | 0086 | 0087 | 0088 | CB31--681F |
| 0089 | 0090 | 0091 | 0092 | CB31--751F |
| 0093 | 0094 | 0095 | 0096 | CB31--102F |
| 0097 | 0098 | 0099 | 0100 | CB31--152F |
| 0101 | 0102 | 0103 | 0104 | CB31--182F |
| 0105 | 0106 | 0107 | 0108 | CB31--222F |
| 0109 | 0110 | 0111 | 0112 | CB31--242F |
| 0113 | 0114 | 0115 | 0116 | CB31--510G |
| 0117 | 0118 | 0119 | 0120 | CB31--560G |
| 0121 | 0122 | 0123 | 0124 | CB31--620G |
| 0125 | 0126 | 0127 | 0128 | CB31--680G |
| 0129 | 0130 | 0131 | 0132 | CB31--750G |
| 0133 | 0134 | 0135 | 0136 | CB31--101G |
| 0137 | 0138 | 0139 | 0140 | CB31--111G |
| 0141 | 0142 | 0143 | 0144 | CB31--151G |
| 0145 | 0146 | 0147 | 0148 | CB31--161G |
| 0149 | 0150 | 0151 | 0152 | CB31--201G |
| 0153 | 0154 | 0155 | 0156 | CB31--221G |
| 0157 | 0158 | 0159 | 0160 | CB31--241G |
| 0161 | 0162 | 0163 | 0164 | CB31--271G |
| 0165 | 0166 | 0167 | 0168 | CB31--301G |
| 0169 | 0170 | 0171 | 0172 | CB31--331G |
| 0173 | 0174 | 0175 | 0176 | CB31--361G |
| 0177 | 0178 | 0179 | 0180 | CB31--431G |
| 0181 | 0182 | 0183 | 0184 | CB31--471G |
| 0185 | 0186 | 0187 | 0188 | CB31--511G |

1/ See footnote at end of table.



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TABLE II. Dash number to part number conversion (continued).

| Dash number in MIL-C-10950/3D | | | | Converted to part number 1/ in MIL-C-10950/3E |
|---|---|---|---|---|
| Characteristic D | | Characteristic E | | |
| Terminal assembly | | | | |
| P | Q | P | Q | |
| 0189 | 0190 | 0191 | 0192 | CB31--561G |
| 0193 | 0194 | 0195 | 0196 | CB31--620G |
| 0197 | 0198 | 0199 | 0200 | CB31--681G |
| 0201 | 0202 | 0203 | 0204 | CB31--750G |
| 0205 | 0206 | 0207 | 0208 | CB31--102G |
| 0209 | 0210 | 0211 | 0212 | CB31--152G |
| 0213 | 0214 | 0215 | 0216 | CB31--182G |
| 0217 | 0218 | 0219 | 0220 | CB31--222G |
| 0221 | 0222 | 0223 | 0224 | CB31--242G |
| 0225 | 0226 | | | CB31--D200J |
| 0227 | 0228 | | | CB31--D220J |
| 0229 | 0230 | | | CB31--D240J |
| 0231 | 0232 | | | CB31--D270J |
| 0233 | 0234 | | | CB31--D300J |
| 0235 | 0236 | | | CB31--D330J |
| 0237 | 0238 | | | CB31--D390J |
| 0239 | 0240 | | | CB31--D470J |
| 0241 | 0242 | 0243 | 0244 | CB31--510J |
| 0245 | 0246 | 0247 | 0248 | CB31--560J |
| 0249 | 0250 | 0251 | 0252 | CB31--620J |
| 0253 | 0254 | 0255 | 0256 | CB31--680J |
| 0257 | 0258 | 0259 | 0260 | CB31--750J |
| 0261 | 0262 | 0263 | 0264 | CB31--101J |
| 0265 | 0266 | 0267 | 0268 | CB31--111J |
| 0269 | 0270 | 0271 | 0272 | CB31--151J |
| 0273 | 0274 | 0275 | 0276 | CB31--161J |
| 0277 | 0278 | 0279 | 0280 | CB31--201J |
| 0281 | 0282 | 0283 | 0284 | CB31--221J |
| 0285 | 0286 | 0287 | 0288 | CB31--241J |
| 0289 | 0290 | 0291 | 0292 | CB31--271J |
| 0293 | 0294 | 0295 | 0296 | CB31--301J |
| 0297 | 0298 | 0299 | 0300 | CB31--331J |
| 0301 | 0302 | 0303 | 0304 | CB31--361J |
| 0305 | 0306 | 0307 | 0308 | CB31--431J |
| 0309 | 0310 | 0311 | 0312 | CB31--471J |
| 0313 | 0314 | 0315 | 0316 | CB31--511J |
| 0317 | 0318 | 0319 | 0320 | CB31--561J |
| 0321 | 0322 | 0323 | 0324 | CB31--621J |
| 0325 | 0326 | 0327 | 0328 | CB31--681J |
| 0329 | 0330 | 0331 | 0332 | CB31--751J |
| 0333 | 0334 | 0335 | 0336 | CB31--102J |
| 0337 | 0338 | 0339 | 0340 | CB31--152J |
| 0341 | 0342 | 0343 | 0344 | CB31--182J |
| 0345 | 0346 | 0347 | 0348 | CB31--222J |
| 0349 | 0350 | 0351 | 0352 | CB31--242J |

1/ See footnote at end of table.



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TABLE II. Dash number to part number conversion (continued).

| Dash number in MIL-C-10950/3D | | | | Converted to part number 1/ in MIL-C-10950/3E |
|---|---|---|---|---|
| Characteristic B | | Characteristic D | | |
| Terminal assembly | | | | |
| P | Q | P | Q | |
| 0353 | 0354 | | | CB31--B050K |
| | | 0355 | 0356 | CB31--D100K |
| | | 0357 | 0358 | CB31--D150K |
| | | 0359 | 0360 | CB31--D180K |
| | | 0361 | 0362 | CB31--D220K |
| | | 0363 | 0364 | CB31--D270K |
| | | 0365 | 0366 | CB31--D330K |
| | | 0367 | 0368 | CB31--D390K |
| | | 0369 | 0370 | CB31--D470K |
| | | 0371 | 0372 | CB31--D560K |
| | | 0373 | 0374 | CB31--D680K |
| | | 0375 | 0376 | CB31--D750K |
| | | 0377 | 0378 | CB31--D101K |
| | | 0379 | 0380 | CB31--D111K |
| | | 0381 | 0382 | CB31--D151K |
| | | 0383 | 0384 | CB31--D161K |
| | | 0385 | 0386 | CB31--D201K |
| | | 0387 | 0388 | CB31--D221K |
| | | 0389 | 0390 | CB31--D271K |
| | | 0391 | 0392 | CB31--D331K |
| | | 0393 | 0394 | CB31--D361K |
| | | 0395 | 0396 | CB31--D431K |
| | | 0397 | 0398 | CB31--D471K |
| | | 0399 | 0400 | CB31--D561K |
| | | 0401 | 0402 | CB31--D681K |
| | | 0403 | 0404 | CB31--D751K |
| | | 0405 | 0406 | CB31--D102K |
| | | 0407 | 0408 | CB31--D152K |
| | | 0409 | 0410 | CB31--D182K |
| | | 0411 | 0412 | CB31--D222K |
| | | 0413 | 0414 | CB31--D242K |

1/ The complete part number shall include symbols for terminal assembly and characteristic, as applicable.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Navy - AS, OS, CG

Preparing activity:
Navy - EC

(Project 5910-0847-4)



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